

Conversions: **C47/R47 Structural and FEA conversions**

Introduction

Structural engineers routinely move between SI and imperial data, and between the intuitive engineering units of a property table and the non-intuitive consistent units an FEA solver demands. The R47/C47 now provides a menu hierarchy under CONV that covers these conversions across sections, materials, loads and mass.

This document describes the Struct: sub-menu, the display modes best suited to structural work, and the number entry and conversion conventions. Two worked examples follow: converting imperial section and material properties to metric equivalents, and preparing consistent-unit inputs for an FEA model.

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Structural engineering parameters

Structural engineering involves calculating strengths of materials and structures. Design of structures often involves following international standards with local code variations relevant to the region. For example, regions subject to snow require compliance with additional snow loads and tropical areas require compliance with high wind loads. Depending on the region, calculations and methods are generally based on SI units or Imperial units. These standard units are often modified with prefixes to generate whole numbers that are easy to display and tabulate.

Tables of material properties are generally available from suppliers, as well as standard industry resources. Depending on the source of the materials, data is often only published in Metric units or Imperial units.

When working with FEA (Finite Element Analysis) systems, the software enforces that a consistent system of units is used. This will often require the conversion of properties to alternative units that are non-intuitive. For example, creating an FEA model in mm, using N for force and kg for mass will require that material density in units of tonnes/mm³. Imperial units are even less intuitive.

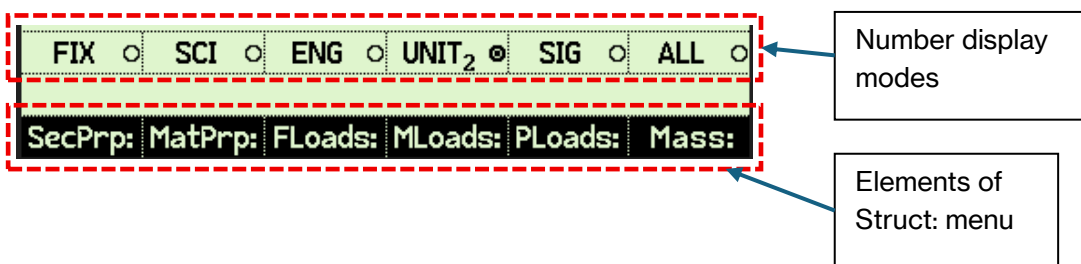
The C47/R47 now provides a Menu hierarchy of conversion functions that cover the structural engineering domain. This menu system is easy to use and navigate.

The menu system is accessed via:

R47: [f, 2] (CONV) C47: [f, 5] (CONV)

Misc:	Struct:				FFF+:
Power:	Ymmv:	Angle:	Speed:	F&p:	Chef:
Energy:	Mass:	Temp:	Length:	Area:	Volume:

[g, F2] (Struct:)



Number display modes

Display mode	Description	Example 12345.6789	Display example
FIX	Fixed point decimal display	FIX-2 (2 decimal places)	12345.68
SCI	Scientific display mode. Exponential display.	SCI-3 (3 decimal places)	1.235×10^4
ENG	Engineering display mode. Exponent is a multiple of 3	ENG-3	12.35×10^3
UNIT	Exponent displayed as SI suffix (eg. m, k, M)	UNIT-3	12.35 k
SIG	Significant digits	SIG-3	12350.
ALL	All decimal places	ALL-3	12345.6789

FIX and UNIT display modes can be most useful with structural conversions.

To make use of the UNIT display mode, number input should be set to Real.

To set Real input mode:

R47: [f , 1] (PREF Menu) [↑, f, F2] (iR)

C47: [f , CHS] (MODE Menu) [↑, f, F2] (iR)

Struct: sub-menu details

Menu	Description	Details																		
SecPrp:	Section property conversions	<table><tr><td>Struct:</td><td>SecPrp:</td><td>$\text{in}^3 \rightarrow \leftarrow \text{mm}^3$</td><td>$\text{in}^6 \rightarrow \leftarrow \text{mm}^6$</td></tr><tr><td></td><td></td><td>$\text{in}^2 \rightarrow \leftarrow \text{mm}^2$</td><td>$\text{in}^4 \rightarrow \leftarrow \text{mm}^4$</td></tr><tr><td>$\text{lbs/ft} \rightarrow \leftarrow \text{kg/m}$</td><td>$\text{in} \rightarrow \leftarrow \text{mm}$</td><td>$\cdot \text{m}$</td><td>$\cdot \text{k}$</td></tr></table> Typical section properties given in property tables for steel and ALU sections. Includes: mass/unit length- (kg/m, lbf/ft), area (mm^2, in^2), Section modulus- (mm^3, in^3), 2nd moment of area- (mm^4, in^4), Warping constant- (mm^6, in^6)	Struct:	SecPrp:	$\text{in}^3 \rightarrow \leftarrow \text{mm}^3$	$\text{in}^6 \rightarrow \leftarrow \text{mm}^6$			$\text{in}^2 \rightarrow \leftarrow \text{mm}^2$	$\text{in}^4 \rightarrow \leftarrow \text{mm}^4$	$\text{lbs/ft} \rightarrow \leftarrow \text{kg/m}$	$\text{in} \rightarrow \leftarrow \text{mm}$	$\cdot \text{m}$	$\cdot \text{k}$						
Struct:	SecPrp:	$\text{in}^3 \rightarrow \leftarrow \text{mm}^3$	$\text{in}^6 \rightarrow \leftarrow \text{mm}^6$																	
		$\text{in}^2 \rightarrow \leftarrow \text{mm}^2$	$\text{in}^4 \rightarrow \leftarrow \text{mm}^4$																	
$\text{lbs/ft} \rightarrow \leftarrow \text{kg/m}$	$\text{in} \rightarrow \leftarrow \text{mm}$	$\cdot \text{m}$	$\cdot \text{k}$																	
MatPrp:	Material property conversions	<table><tr><td>Struct:</td><td>MatPrp:</td><td>$\text{lbs/in}^3 \rightarrow \leftarrow \text{blob/in}^3$</td><td>$\text{kg/m}^3 \rightarrow \leftarrow \text{blob/in}^3$</td></tr><tr><td>$\text{ksi} \rightarrow \leftarrow \text{MPa}$</td><td>$\text{lbs/in}^3 \rightarrow \leftarrow \text{t/mm}^3$</td><td>$\text{kg/m}^3 \rightarrow \leftarrow \text{t/mm}^3$</td><td></td></tr><tr><td>$\text{psi} \rightarrow \leftarrow \text{Pa}$</td><td>$\text{lbs/in}^3 \rightarrow \leftarrow \text{kg/m}^3$</td><td>$\cdot \text{m}$</td><td>$\cdot \text{k}$</td></tr></table> Typical material properties. Includes modulus of elasticity, Yield strength, Ultimate strength (Pa, psi), Density ($\text{kg/m}^3, \text{t/mm}^3, \text{lbs/in}^3, \text{blobs/in}^3$)	Struct:	MatPrp:	$\text{lbs/in}^3 \rightarrow \leftarrow \text{blob/in}^3$	$\text{kg/m}^3 \rightarrow \leftarrow \text{blob/in}^3$	$\text{ksi} \rightarrow \leftarrow \text{MPa}$	$\text{lbs/in}^3 \rightarrow \leftarrow \text{t/mm}^3$	$\text{kg/m}^3 \rightarrow \leftarrow \text{t/mm}^3$		$\text{psi} \rightarrow \leftarrow \text{Pa}$	$\text{lbs/in}^3 \rightarrow \leftarrow \text{kg/m}^3$	$\cdot \text{m}$	$\cdot \text{k}$						
Struct:	MatPrp:	$\text{lbs/in}^3 \rightarrow \leftarrow \text{blob/in}^3$	$\text{kg/m}^3 \rightarrow \leftarrow \text{blob/in}^3$																	
$\text{ksi} \rightarrow \leftarrow \text{MPa}$	$\text{lbs/in}^3 \rightarrow \leftarrow \text{t/mm}^3$	$\text{kg/m}^3 \rightarrow \leftarrow \text{t/mm}^3$																		
$\text{psi} \rightarrow \leftarrow \text{Pa}$	$\text{lbs/in}^3 \rightarrow \leftarrow \text{kg/m}^3$	$\cdot \text{m}$	$\cdot \text{k}$																	
FLoads:	Force-based load conversions	<table><tr><td>Struct:</td><td>FLoads:</td><td>$\text{kgf} \rightarrow \leftarrow \text{N}$</td><td>$\text{m/s}^2 \rightarrow \leftarrow \text{ft/s}^2$</td></tr><tr><td>$\text{kgf/m} \rightarrow \leftarrow \text{N/m}$</td><td>$\text{lbf} \rightarrow \leftarrow \text{kgf}$</td><td>$\text{m/s}^2 \rightarrow \leftarrow \text{in/s}^2$</td><td></td></tr><tr><td>$\text{lbf/ft} \rightarrow \leftarrow \text{N/m}$</td><td>$\text{lbf} \rightarrow \leftarrow \text{N}$</td><td>$\cdot \text{m}$</td><td>$\cdot \text{k}$</td></tr></table> Typical force-based loads. Includes force/length (N/m, kg/m, lbs/ft), force (kgf, N, lbf), acceleration, gravity ($\text{m/s}^2, \text{ft/s}^2, \text{in/s}^2$)	Struct:	FLoads:	$\text{kgf} \rightarrow \leftarrow \text{N}$	$\text{m/s}^2 \rightarrow \leftarrow \text{ft/s}^2$	$\text{kgf/m} \rightarrow \leftarrow \text{N/m}$	$\text{lbf} \rightarrow \leftarrow \text{kgf}$	$\text{m/s}^2 \rightarrow \leftarrow \text{in/s}^2$		$\text{lbf/ft} \rightarrow \leftarrow \text{N/m}$	$\text{lbf} \rightarrow \leftarrow \text{N}$	$\cdot \text{m}$	$\cdot \text{k}$						
Struct:	FLoads:	$\text{kgf} \rightarrow \leftarrow \text{N}$	$\text{m/s}^2 \rightarrow \leftarrow \text{ft/s}^2$																	
$\text{kgf/m} \rightarrow \leftarrow \text{N/m}$	$\text{lbf} \rightarrow \leftarrow \text{kgf}$	$\text{m/s}^2 \rightarrow \leftarrow \text{in/s}^2$																		
$\text{lbf/ft} \rightarrow \leftarrow \text{N/m}$	$\text{lbf} \rightarrow \leftarrow \text{N}$	$\cdot \text{m}$	$\cdot \text{k}$																	
MLoads:	Moment-based load conversion	<table><tr><td>Struct:</td><td>MLoads:</td><td>FIX₂</td><td>SCI</td><td>ENG</td><td>UNIT</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>$\text{ft}\cdot\text{lbf} \rightarrow \leftarrow \text{N}\cdot\text{m}$</td><td>$\text{in}\cdot\text{lbf} \rightarrow \leftarrow \text{N}\cdot\text{m}$</td><td>$\cdot \text{m}$</td><td>$\cdot \text{k}$</td><td></td><td></td></tr></table> Typical moment-based loads Includes Moments (N.m, ft.lbf, in.lbf)	Struct:	MLoads:	FIX₂	SCI	ENG	UNIT							$\text{ft}\cdot\text{lbf} \rightarrow \leftarrow \text{N}\cdot\text{m}$	$\text{in}\cdot\text{lbf} \rightarrow \leftarrow \text{N}\cdot\text{m}$	$\cdot \text{m}$	$\cdot \text{k}$		
Struct:	MLoads:	FIX₂	SCI	ENG	UNIT															
$\text{ft}\cdot\text{lbf} \rightarrow \leftarrow \text{N}\cdot\text{m}$	$\text{in}\cdot\text{lbf} \rightarrow \leftarrow \text{N}\cdot\text{m}$	$\cdot \text{m}$	$\cdot \text{k}$																	

Menu	Description	Details
PLoads:	Pressure-base load conversions	Struct: PLoads: FIX₂ SCI ENG UNIT lbf/ft² → ← Pa psi → ← Pa -m -k Typical pressure loads Force/unit area (Pa, lbf/ft², psi)
Mass:	Mass conversions	slug → ← kg slinch → ← kg blob → ← kg stone → ← kg short cwt → ← kg tr oz → ← g lbs → ← kg cwt → ← kg oz → ← g lbf.s²/ft → ← kg lbf.s²/in → ← kg lbf.s²/in → ← kg carat → ← g jin → ← kg liang → ← kg long ton → ← kg short ton → ← kg tonne → ← kg Wide variety of mass conversions. Includes (kg, lbs, tonnes, tons, short tons, slugs, lbf.s²/ft, slinch, blob, lbf.s²/in.

General menu structure

The diagram shows a menu structure with two main sections: 'Struct:' and 'SecPrp:'. The 'Struct:' section includes units like in³, in⁶, in², in⁴, lbs/ft, and in. The 'SecPrp:' section includes units like mm³, mm⁶, mm², mm⁴, kg/m, and mm. Below the menu structure, there are three callouts: 'Navigate to previous menu' pointing to the left arrow, 'Current menu name' pointing to the 'SecPrp:' label, and 'Prefix operators: m – milli – multiply by 10⁻³, k – kilo – multiply by 10³. (These are a subset of operators available under the PFX menu [f, EEX]). These are very helpful for entering larger values.' pointing to the '-m' and '-k' operators.

Number entry examples

With one of the Struct: sub-menus active

Display mode FIX-2 (iR set)

To enter 135,000: [130, F6] Display: 135 000
 To enter 124,000,000: [124 F6, F6] Display: 124 000 000
 To enter 1.24 x 10⁻³: [1.24 F5] Display: 1.24 × (10)⁻³

Display mode UNIT-2 (iR set)

To enter 135,000: [130, F6] Display: 135. k
 To enter 124,000,000: [124 F6, F6] Display: 124. M

To enter 1.24×10^{-3} : [1.24 F5]

Display: 1.24 m

Conversion direction arrows

The **CONVhp** flag is used to control the direction of the arrows in the conversion menus. This provides a great visual clue when performing conversions.

CONVhp cleared:

Usage: I HAVE in^3 and WANT mm^3

Struct:	SecPrp:	$in^3 \rightarrow \leftarrow mm^3$	$in^6 \rightarrow \leftarrow mm^6$
		$in^2 \rightarrow \leftarrow mm^2$	$in^4 \rightarrow \leftarrow mm^4$
$lbs/ft \rightarrow \leftarrow kg/m$		$in \rightarrow \leftarrow mm$	$\cdot m \quad \cdot k$

CONVhp set:

Usage: I WANT mm^3 and HAVE in^3

Struct:	SecPrp:	$mm^3 \leftarrow \rightarrow in^3$	$mm^6 \leftarrow \rightarrow in^6$
		$mm^2 \leftarrow \rightarrow in^2$	$mm^4 \leftarrow \rightarrow in^4$
$kg/m \leftarrow \rightarrow lbs/ft$		$mm \leftarrow \rightarrow in$	$\cdot m \quad \cdot k$

Conversion result unit display

The displays also provide feedback after a conversion is done to give visual verification that the correct conversion has been performed. See below after performing a conversion of in^4 to mm^4

2026-07-02
L ° /64x
64:2

0.00
0.00
0.00

mm⁴:

156 086 784.60

Struct:	SecPrp:	$in^3 \rightarrow \leftarrow mm^3$	$in^6 \rightarrow \leftarrow mm^6$
		$in^2 \rightarrow \leftarrow mm^2$	$in^4 \rightarrow \leftarrow mm^4$
$lbs/ft \rightarrow \leftarrow kg/m$		$in \rightarrow \leftarrow mm$	$\cdot m \quad \cdot k$

Conversion examples

Conversion example 1:

You need to manufacture a structure that has been designed using steel sections available in the US (imperial dimensions). You want to use alternative sections that are available in your country (metric sections).

US section properties:

W16x31 wide flange beam

$I_{xx} = 375 in^4$

Sx: = 47.2 in^3

Yield strength: 36 ksi (A36 grade)

Calculate equivalent metric properties

Action	Keystrokes	Display
Display SecPrp: menu	R47: [f, 2, g, F2, F1] C47: [f, 5, g, F2, F1]	Struct: SecPrp: $\text{in}^3 \rightarrow \leftarrow \text{mm}^3$ $\text{in}^6 \rightarrow \leftarrow \text{mm}^6$ $\text{in}^2 \rightarrow \leftarrow \text{mm}^2$ $\text{in}^4 \rightarrow \leftarrow \text{mm}^4$ lbs/ft $\rightarrow \leftarrow$ kg/m $\text{in} \rightarrow \leftarrow \text{mm}$ $\cdot \text{m}$ $\cdot \text{k}$
Convert Ixx	[375, f, F5]	mm^4: 156 086 784.60 Struct: SecPrp: $\text{in}^3 \rightarrow \leftarrow \text{mm}^3$ $\text{in}^6 \rightarrow \leftarrow \text{mm}^6$ $\text{in}^2 \rightarrow \leftarrow \text{mm}^2$ $\text{in}^4 \rightarrow \leftarrow \text{mm}^4$ lbs/ft $\rightarrow \leftarrow$ kg/m $\text{in} \rightarrow \leftarrow \text{mm}$ $\cdot \text{m}$ $\cdot \text{k}$
Convert Sx	[47.3, g, F3]	mm^3: 775 108.13 Struct: SecPrp: $\text{in}^3 \rightarrow \leftarrow \text{mm}^3$ $\text{in}^6 \rightarrow \leftarrow \text{mm}^6$ $\text{in}^2 \rightarrow \leftarrow \text{mm}^2$ $\text{in}^4 \rightarrow \leftarrow \text{mm}^4$ lbs/ft $\rightarrow \leftarrow$ kg/m $\text{in} \rightarrow \leftarrow \text{mm}$ $\cdot \text{m}$ $\cdot \text{k}$
Display MatPrp: menu	[g, F1, F2]	Struct: MatPrp: $\text{lbs/in}^3 \rightarrow \leftarrow \text{blob/in}^3$ $\text{kg/m}^3 \rightarrow \leftarrow \text{blob/in}^3$ ksi $\rightarrow \leftarrow$ MPa $\text{lbs/in}^3 \rightarrow \leftarrow \text{t/mm}^3$ $\text{kg/m}^3 \rightarrow \leftarrow \text{t/mm}^3$ psi $\rightarrow \leftarrow$ Pa $\text{lbs/in}^3 \rightarrow \leftarrow \text{kg/m}^3$ $\cdot \text{m}$ $\cdot \text{k}$
Convert yield strength	[36, f, F1]	MPa: 248.21 Struct: MatPrp: $\text{lbs/in}^3 \rightarrow \leftarrow \text{blob/in}^3$ $\text{kg/m}^3 \rightarrow \leftarrow \text{blob/in}^3$ ksi $\rightarrow \leftarrow$ MPa $\text{lbs/in}^3 \rightarrow \leftarrow \text{t/mm}^3$ $\text{kg/m}^3 \rightarrow \leftarrow \text{t/mm}^3$ psi $\rightarrow \leftarrow$ Pa $\text{lbs/in}^3 \rightarrow \leftarrow \text{kg/m}^3$ $\cdot \text{m}$ $\cdot \text{k}$

From Table 9 below for Yield strength 300MPa steel, 360UB 56.7 is the smallest beam that provides the necessary properties to replace the US sourced beam.

Table 9 Universal Beams – Dimensions and Properties

Designation	Depth of Section	Flange		Web Thickness	Root Radius	Depth Between Flanges			Gross Area of Cross Section	About x-axis				
		Width	Thickness				d_i	$(b_f t_w)$		A_g	I_x	Z_x	S_x	r_x
	d	b_f	t_f	t_w	r_1	d_i	t_w	$2t_f$	A_g	I_x	Z_x	S_x	r_x	
	kg/m	mm	mm	mm	mm	mm			mm ²	10 ⁶ mm ⁴	10 ³ mm ³	10 ³ mm ³	mm	
610 UB 125	611.6	229.0	19.6	11.9	14.0	572.4	48.1	5.54	16000	986	3230	3680	249	
113	607.0	228.0	17.3	11.2	14.0	572.4	51.1	6.27	14500	875	2880	3290	246	
101	602.0	228.0	14.8	10.6	14.0	572.4	54.0	7.34	13000	761	2530	2900	242	
530 UB 92.4	533.0	209.0	15.6	10.2	14.0	501.8	49.2	6.37	11800	554	2080	2370	217	
82.0	528.2	209.0	13.2	9.6	14.0	501.8	52.3	7.55	10500	477	1810	2070	213	
460 UB 82.1	460.4	191.0	16.0	9.9	11.4	428.4	43.3	5.66	10500	372	1610	1840	188	
74.6	457.4	190.0	14.5	9.1	11.4	428.4	47.1	6.24	9520	335	1460	1660	188	
67.1	453.8	190.0	12.7	8.5	11.4	428.4	50.4	7.15	8580	296	1300	1480	186	
410 UB 59.7	406.4	178.0	12.8	7.8	11.4	380.8	48.8	6.65	7640	216	1060	1200	168	
53.7	402.6	178.0	10.9	7.6	11.4	380.8	50.1	7.82	6890	188	933	1060	165	
360 UB 56.7	358.6	172.0	13.0	8.0	11.4	332.6	41.6	6.31	7240	161	899	1010	149	
50.7	355.6	171.0	11.5	7.3	11.4	332.6	45.6	7.12	6470	142	798	897	148	
44.7	352.0	171.0	9.7	6.9	11.4	332.6	48.2	8.46	5720	121	689	777	146	
310 UB 46.2	307.2	166.0	11.8	6.7	11.4	283.6	42.3	6.75	5930	100	654	729	130	
40.4	304.0	165.0	10.2	6.1	11.4	283.6	46.5	7.79	5210	86.4	569	633	129	
32.0	298.0	149.0	8.0	5.5	13.0	282.0	51.3	8.97	4080	63.2	424	475	124	
250 UB 37.3	256.2	146.0	10.9	6.4	8.9	234.4	36.6	6.40	4750	55.7	435	486	108	
31.4	251.6	146.0	8.6	6.1	8.9	234.4	38.4	8.13	4010	44.5	354	397	105	
25.7	248.0	124.0	8.0	5.0	12.0	232.0	46.4	7.44	3270	35.4	285	319	104	
200 UB 29.8	207.0	134.0	9.6	6.3	8.9	187.8	29.8	6.65	3820	29.1	281	316	87.3	
25.4	203.2	133.0	7.8	5.8	8.9	187.6	32.3	8.15	3230	23.6	232	260	85.4	
22.3	201.6	133.0	7.0	5.0	8.9	187.6	37.5	9.14	2870	21.0	208	231	85.5	
18.2	198.0	99.0	7.0	4.5	11.0	184.0	40.9	6.75	2320	15.8	160	180	82.6	

Conversion example 2:

You are performing a FEA analysis on an ALU part. The FEA is using units of inch, lbs. You need to provide several parameters for the FEA configuration using ALU grade 6061-T651:

ALU Properties:

Modulus of elasticity: 68 GPa

Yield strength: 245 MPa

Density: 2700 kg/m³

Load case:

Gravity load of 3.5g

Calculate equivalent material properties for FEA input

Action	Keystrokes	Display
Display MatPrp: menu	R47: [f, 2, g, F2, F2] C47: [f, 5, g, F2, F2]	Struct: MatPrp: lbs/in³ → ← blob/in³ kg/m³ → ← blob/in³ ksi → ← MPa lbs/in³ → ← t/mm³ kg/m³ → ← t/mm³ psi → ← Pa lbs/in³ → ← kg/m³ ·m ·k

Action	Keystrokes	Display
Convert Modulus of elasticity to units of psi	[68, EEX, 9 ENTER, F2]	psi: 9 862 566.17 Struct: MatPrp: lbs/in³ → ← blob/in³ kg/m³ → ← blob/in³ ksi → ← MPa lbs/in³ → ← t/mm³ kg/m³ → ← t/mm³ psi → ← Pa lbs/in³ → ← kg/m³ ·m ·k
Convert Yield strength to units of ksi	[245, f, F2]	ksi: 35.53 Struct: MatPrp: lbs/in³ → ← blob/in³ kg/m³ → ← blob/in³ ksi → ← MPa lbs/in³ → ← t/mm³ kg/m³ → ← t/mm³ psi → ← Pa lbs/in³ → ← kg/m³ ·m ·k
Display FLoads menu	[g, F1, F3]	Struct: FLoads: kgf → ← N m/s² → ← ft/s² kgf/m → ← N/m lbf → ← kgf m/s² → ← in/s² lbf/ft → ← N/m lbf → ← N ·m ·k
Convert gravity load to units of in/s ²	[3.5, ENTER, 9.8, ×, f, F5]	in/s²: 385.83 Struct: FLoads: kgf → ← N m/s² → ← ft/s² kgf/m → ← N/m lbf → ← kgf m/s² → ← in/s² lbf/ft → ← N/m lbf → ← N ·m ·k